

120 Days of Quant Finance

Week 2 – Statistical Inference

Day 2: Hypothesis Testing and p-values

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Introduction

Statistical inference concerns decision-making under uncertainty. Hypothesis testing provides a formal framework for assessing whether observed data is compatible with a proposed probabilistic model.

Rather than proving statements to be true or false, hypothesis testing evaluates whether the observed data provides sufficient evidence to question a default assumption.

Statistical Hypotheses

A hypothesis test begins with the formulation of two competing hypotheses.

Null Hypothesis

The **null hypothesis**, denoted by H_0 , represents the default or baseline assumption. It typically corresponds to the absence of an effect or deviation.

A common example is:

$$H_0 : \mu = \mu_0,$$

where μ is an unknown population parameter.

Alternative Hypothesis

The **alternative hypothesis**, denoted by H_1 , represents a deviation from the null hypothesis. Depending on the context, it may be two-sided or one-sided:

$$H_1 : \mu \neq \mu_0, \quad H_1 : \mu > \mu_0, \quad H_1 : \mu < \mu_0.$$

In hypothesis testing, evidence is evaluated against the null hypothesis.

Test Statistics

A **test statistic** is a function of the observed data designed to measure the degree of departure from the null hypothesis.

Consider a sample $X_1, \dots, X_n$ with sample mean $\bar{X}$, known population standard deviation σ , and sample size n . The standardized test statistic is given by:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

By the Central Limit Theorem, under H_0 :

$$Z \sim \mathcal{N}(0, 1).$$

This sampling distribution forms the basis for computing probabilities under the null hypothesis.

Sampling Distribution Under the Null

Hypothesis testing evaluates how extreme the observed value of the test statistic is relative to its distribution under the assumption that H_0 is true.

Extremeness is defined in terms of tail probabilities of the sampling distribution.

Definition of the p-value

The **p-value** is defined as the probability, computed under the null hypothesis, of observing a test statistic at least as extreme as the one obtained from the data.

For a two-sided test:

$$\text{p-value} = P(|Z| \geq |z_{\text{obs}}| \mid H_0).$$

For a one-sided test:

$$\text{p-value} = P(Z \geq z_{\text{obs}} \mid H_0) \quad \text{or} \quad P(Z \leq z_{\text{obs}} \mid H_0).$$

The p-value quantifies the compatibility between the observed data and the null hypothesis.

Interpretation of the p-value

The p-value should be interpreted carefully:

- It is the probability of the observed data, or more extreme outcomes, assuming H_0 is true.
- It is not the probability that H_0 is true.
- It does not measure the magnitude or importance of an effect.

Misinterpretation of p-values is a common source of statistical error.

Significance Level and Decision Rule

A hypothesis test requires the specification of a significance level $\alpha \in (0, 1)$, commonly chosen as $\alpha = 0.05$.

The decision rule is:

- Reject H_0 if p-value $\leq \alpha$.
- Fail to reject H_0 if p-value $> \alpha$.

The significance level represents the maximum acceptable probability of a Type I error.

Type I and Type II Errors

Two types of errors may occur in hypothesis testing:

- **Type I error:** rejecting H_0 when it is true, with probability α .
- **Type II error:** failing to reject H_0 when it is false, with probability β .

The power of a test is defined as:

$$\text{Power} = 1 - \beta.$$

Power measures the ability of a test to detect deviations from the null hypothesis.

One-Sided and Two-Sided Tests

In a two-sided test, deviations in both directions are considered:

$$H_1 : \mu \neq \mu_0.$$

In a one-sided test, deviations in only one direction are considered:

$$H_1 : \mu > \mu_0 \quad \text{or} \quad H_1 : \mu < \mu_0.$$

The choice between one-sided and two-sided tests must be determined before observing the data.

Connection to the Central Limit Theorem

The Central Limit Theorem provides the theoretical justification for approximating the sampling distribution of test statistics by the Normal distribution.

Without the CLT, classical hypothesis testing procedures would not be valid in many practical settings.

Conclusion

Hypothesis testing provides a structured approach to decision-making under uncertainty by quantifying the evidence against a null hypothesis.

A correct understanding of test statistics, p-values, and error probabilities is essential for responsible statistical inference and for interpreting results in applied settings.